Classification of Post-Stroke Motor Imagery Using Per-Channel Fractal EEG Features: Conclusions

Niveles3 — Fractal EEG Classification Pipeline

1. Per-electrode fractal features with a standard SVM achieve 85.46% three-class accuracy at Month 6 post-stroke (chronic phase). The combination of four complementary fractal methods (RS, Higuchi, DFA, Semivariogram) across 16 channels (64 features) with SMOTE and an RBF-kernel SVM ($C=12$, $\gamma=0.02$) outperformed all other configurations (+17.7 pts over baseline).

2. Accuracy shows a progressive temporal trajectory correlating with neurological recovery. SVM evolved from 75.0% (Month 1) to 84.3% (Month 3) to 85.5% (Month 6). This convergence was consistent across all metrics (κ: +0.625 → +0.782; MAE: 7.50 → 4.50).

3. Higuchi's fractal dimension is the single most discriminative feature. Evaluated alone (16 features), Higuchi reached 81.2% (Month 6) and 83.5% (Month 3), outperforming DFA (77.0%), Semivariogram (78.0%), and RS (68.0%) by 3-13 points. At Month 3, Higuchi alone fell within 0.8 points of the four-method combination, yet the combined set was necessary to maximize MVR 40% detection (TPR 89% vs 78%).

4. The three Hurst exponent methods (HO, HRS, HV) degrade performance when added to the four basic fractal methods. The seven-method set (112 features) achieved 82.4% vs 85.5% (-3.1 pts), with rest-class TPR collapsing from 39% to 22%. This contradicts findings in healthy subjects (Martinez-Peon et al., 2024) and suggests the Hurst exponent is insufficient to capture the multifractal dynamics of post-stroke EEG.

5. Feature quality outweighs feature quantity. The four-method set (64 features) consistently outperformed the seven-method set (112 features). Per-electrode features outperformed spatial-mean aggregation by 15-28 points, confirming that topographic information is critical.

6. Inter-patient variability remains the principal obstacle to clinical deployment. Per-patient accuracy at Month 6 ranged from 77.8% (P1) to 90.5% (P3), with a standard deviation of ±5.1 pts. Leave-one-patient-out (LOPO) inter-subject evaluation reached only 63.65%, underscoring the need for patient-specific calibration.

KEY METRICS (SVM, 3-class, Month 6): Accuracy 85.46% | Kappa +0.749 | MAE 4.50
TPR Rest 38.8% | TPR MVR10% 91.3% | TPR MVR40% 88.8%